

HW3 Part 2 — RIT ALGO2 Market Making Write-up

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Setup and economics

ALGO2 rewards passive liquidity: a filled limit order earns about 1¢ per share against the mid, while a market order costs about 1.5¢.

The case has one stock (ALGO) traded for 300 seconds. Orders are capped at 5,000 shares, and positions at 25,000 shares gross or net, with a 10¢-per-share fine beyond that. Market orders pay a 1¢ commission; filled limit orders receive a ½¢ rebate.

At the usual 1¢ spread, a passive fill earns half the spread (0.5¢) plus the rebate (0.5¢). A market order pays half the spread plus the commission (1.5¢). Two passive fills in a round trip earn 2¢ per share, and even a round trip with no price gain earns 1¢ in rebates. The goal is therefore to trade passively, as often as possible, while keeping inventory small.

(a) Strategy going in

I went in with v5: a two-sided market maker quoting at the best bid and ask, skewing its quotes against its inventory, and trading 2,500 shares per order.

I built it in versions following the brief's two rules: always keep a bid and an ask near the top of the book, resetting when one leg fills (rule 1); and keep inventory close to balanced (rule 2). Each version was tested in practice cases.

Version	What it added	What the test showed
v1	The brief literally: bid at LAST – 5¢, ask at LAST + 5¢	Orders posted twice and never moved; one leg sat unfilled for 96 ticks
v2	Seven fixes: side-aware state, wait for confirmation, a requote rule, the mid instead of LAST, logging	Sound mechanics
v3	Quotes at the best price: join when the spread is under 3¢, improve by 1¢ otherwise	Frequent fills, but long 13,000 shares after a 5¢ drop: no inventory control

Version	What it added	What the test showed
v4	Inventory skew of 1¢ per 1,000 shares, soft limit of 10,000, passive unwind from tick 280	Unwound 12,000 shares passively; \$3.8 per tick
v5	2,500 shares per order; quotes near the mid when the spread is 4¢ or wider	\$3.9 per tick over two cases
v6	v5 plus a step back on the exposed quote when book imbalance exceeds 0.6	\$2.5 per tick: the guard cost fills

Three choices held across versions: the mid, not LAST, as reference price; quotes never crossing the spread; and moving only the leg that is off target. Market orders were used only to flatten in the last 5 ticks.

I chose v5 because it earned as much per tick as v4 over a larger sample (212 ticks, two cases), with positions within $\pm 2,500$. The practice cases ended at +\$898 and +\$910.

(b) What happened, size strategy and script

The graded case was a trend day: the price rose 125¢ by tick 160, then fell 122¢, and my algorithm held the wrong side of both moves.

What happened

In practice the price stayed within a few cents of 20.00. Here it went from 20.01 to 21.26 by tick 160, then back to 20.04 by tick 300: about 0.8¢ per tick, almost without pause.

I was short 85% of the time during the rise and long 89% of the time during the fall. The largest loss was a 2,500-share short held from tick 23 to 132 while the price rose 96¢, with zero fills: −\$2,400.

Four mechanisms explain it:

1. **Adverse selection.** A rally is buyers taking offers, so the quote that fills is my ask, which leaves me short just before the price rises further.
2. **No passive exit.** My bid to cover sat at the best price 85–90% of the time, but each new price level sent it to the back of the queue. It was reposted 86 times without a fill.
3. **The skew was only a nudge.** It moved my quotes 2¢; the price moved 125¢.
4. **The reset rule re-entered the trend.** After covering at tick 132, the fresh pair's ask was taken again: I went short five more times by tick 147.



Price with my position shaded (red short, green long), my position, and P&L split into the spread side (green) and inventory (red). Dashed lines mark each restart.

Size strategy

In practice, P&L was proportional to volume: about 0.85¢ per share traded. I raised the order size from 1,000 shares (v3-v4) to 2,500 (v5-v6), half the 5,000 cap, to trade more per fill.

Volume did not rise: it stayed near 500 shares per tick, because after each fill the skew pushed one quote off the best price. Fills became 2.5 times rarer. Size is also exposure: during the fall, v4's 1,000-share orders lost about \$9 per tick against about \$21 for v5. The maximum position was 5,198 shares, far from the 10,000 soft limit.

The script

`algo2_mm.py` runs one loop several times per second through the RIT REST API:

1. Read the book, excluding my own orders.
2. Compute the target bid and ask from the best prices and my inventory.

3. No orders: post a pair. One leg filled: cancel the other and reset. Otherwise, move only the leg that is off target.
4. Apply the inventory rules: soft limit, unwind from tick 280, flatten in the last 5 ticks.
5. Log the state and the event to a CSV file.

Usage is in the appendix; the code is attached.

(c) Market orders versus limit orders

Limit orders earn the spread but give up control of execution; market orders guarantee execution but pay for it. A market maker earns with the first and controls risk with the second. My algorithm only used the first.

	Limit order (passive)	Market order (aggressive)
Per share, against the mid	+0.5¢ half-spread +0.5¢ rebate	−0.5¢ half-spread −1¢ commission
Execution	Uncertain: depends on queue position and on the price coming to you	Immediate and certain
Price	Known in advance	Can walk the book if the size exceeds the best level
Main risks	Adverse selection (filled just before the price moves against you), inventory held while waiting, queue priority lost at every cancel and repost	The cost, and slippage on large size
In my graded case	Every quote: +\$425 of rebates, while adverse selection took back the half-spread	None: even the end-of-case unwind filled passively

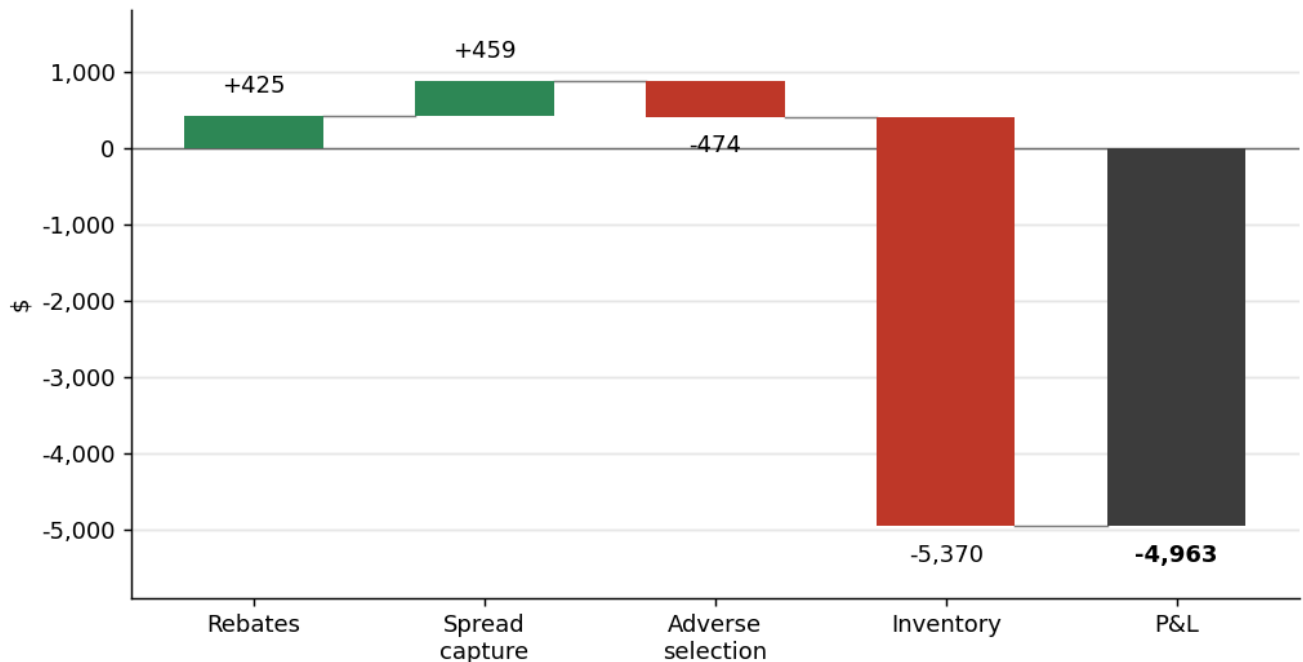
The stuck short shows both sides. My passive bid never filled in 109 ticks. A market order when the price had moved 4¢ against me (tick 33) would have cost about \$137, \$100 of price move plus \$37 of fees and spread, instead of \$2,400.

(d) P&L

My P&L for the graded case was **−\$4,962.74**, flat at the end with no fines. Rebates were the only clean income; inventory caused the loss.

I split the P&L into four drivers, rebuilding each fill's price from my logs and valuing positions at the mid:

- **Rebates (+\$425):** $\frac{1}{2}\text{¢}$ on each of the 85,000 shares filled passively.
- **Spread capture (+\$459):** fill price against the mid just before the fill, about half the 1¢ spread per share.
- **Adverse selection (-\$474):** the mid moving against me during the fill itself. It cancelled the spread capture: I was mostly filled as the price ran through my quote.
- **Inventory (-\$5,370):** positions held while the price moved.



The four drivers sum to -\$4,960, within \$3 of RIT's figure; the gap comes from rebuilding fill prices.

Run	Version	Ticks	Rebates	Spread net of adverse selection	Inventory	Total
1	v5	1-53	+\$88	+\$88	-\$650	-\$475
2	v4	58-73	\$0	\$0	-\$350	-\$350
3	v5	75-89	\$0	\$0	-\$300	-\$300
4	v6	90-161	+\$100	+\$37	-\$1,462	-\$1,325
5	v4	165-224	+\$120	+\$45	-\$695	-\$530
6	v5	228-279	+\$74	+\$12	-\$1,133	-\$1,048
7	v5	288-300	+\$26	+\$26	-\$210	-\$158
Between runs	no script		+\$18	-\$223	-\$570	-\$775
Total		1-300	+\$425	-\$15	-\$5,370	-\$4,960

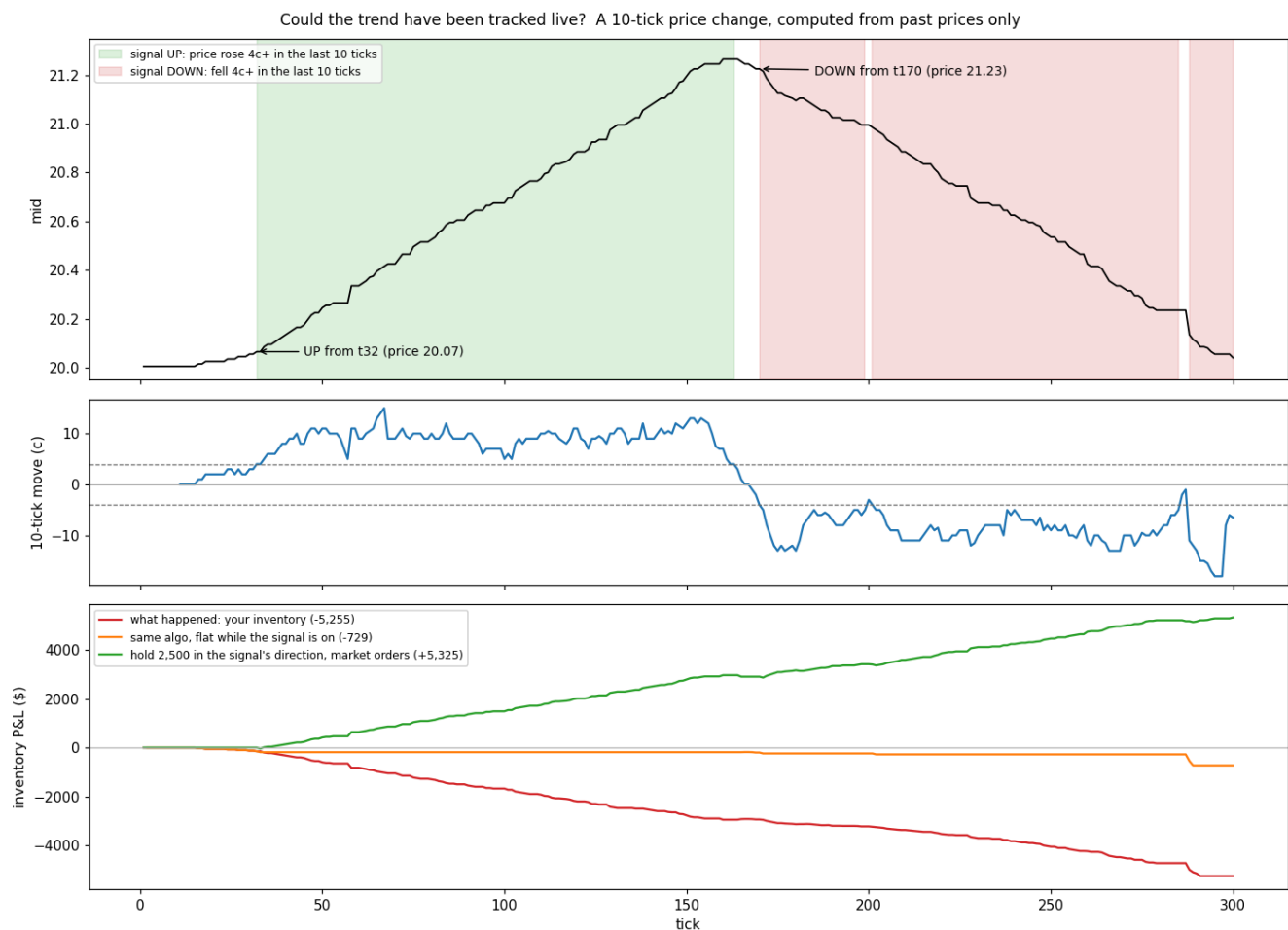
I restarted the algorithm six times, switching versions to adapt as the trend developed: v4 trades smaller orders (1,000 shares), v6 adds the imbalance guard, and v5 is the baseline. Every run shows the same pattern: positive rebates, a spread roughly cancelled by adverse selection, and a large inventory loss. All versions share the same passive core, so none could exit a position going against the trend; smaller orders only slowed the loss.

The restarts themselves cost \$775: positions held while no script was running, and two orders left in the book that filled as the price ran through them. In the two practice cases, where the price stayed within a few cents of 20.00, the same algorithm made +\$898 and +\$910.

What I would do differently

The trend was visible in real time: a simple 10-tick price change flagged it within a few ticks, and standing aside would have cut the inventory loss by about 85%.

- **Track the price.** Trend = mid now minus mid 10 ticks ago; $\pm 4\text{¢}$ or more counts as a trend. On this case it signalled "rising" from tick 32 (price 20.07) and "falling" from tick 170 (21.23). It was on for 98% of both moves and never pointed the wrong way.
- **Stand aside in a trend.** Stop quoting the side the trend runs into, close inventory held against it with a market order, and resume two-sided quoting once the 10-tick move falls under 2¢ .
- **Stop every position.** Close at market if the price moves 4¢ against the average entry price, trend or not.
- **Run one version and watch it.** Keep the same script for the whole case, stop it with Ctrl+C so it cancels its orders, and follow a live monitor that splits P&L into spread and inventory.



Top: the signal computed from past prices only. Bottom: inventory P&L as it happened, if the algorithm had stood aside during trends, and if it had held 2,500 shares with the trend, replayed tick by tick.

The thresholds were chosen after seeing this case, so the 85% is optimistic. In the practice cases the price rarely moved 4¢ in 10 ticks, so the signal would have stayed mostly off and market making would have run as before.

Appendix: how to use the scripts

Both scripts run in the Anaconda Prompt on the course VM, with the RIT client open and its RESTAPI enabled. The API key and port are set at the top of `algo2_mm.py`.

- 1. Start before the case:** `python algo2_mm.py v5`. The script waits for the case to start, trades until tick 300, and writes a CSV log per run plus a summary line in the `logs` folder.
- 2. Other versions:** replace `v5` with `v3`, `v4` or `v6`. `python algo2_mm.py auto v5 v6` splits one case evenly between versions, for testing.
- 3. Stop with Ctrl+C:** the script cancels its open orders before exiting.

4. **Monitor:** `python analyze_log.py --latest` prints a report (P&L split into rebates, spread capture, adverse selection and inventory; fills; cancels; worst inventory episodes) and saves two charts. With `--merge` and several log files, it reads them as one case, with one line per run. Add `--live` to refresh every 5 seconds in a second window during the case.

Setting	v5 value	Role
<code>QTY</code>	2,500 shares	Size of each order
<code>SKEW</code>	1¢ per 1,000 shares	Shifts both quotes against the position
<code>SOFT_LIMIT</code>	10,000 shares	Beyond it, only the side that reduces the position is quoted
<code>UNWIND_BEFORE</code>	20 ticks	From tick 280, only the reducing side, passively
<code>FLATTEN_BEFORE</code>	5 ticks	From tick 295, market orders to finish flat
<code>DEEP</code>	On	Quotes near the mid when the spread is 4¢ or wider
<code>IMB_TH</code>	Off (0.60 in v6)	Steps back the exposed quote when book imbalance is high